Managed ArgoCD Scalability

Final Slide Deck Content

Slide 1

Managed ArgoCD Scalability

Expanding GitOps as a Platform Capability

Objective: Evaluate how Managed ArgoCD can evolve to support broader platform adoption while maintaining a scalable and reliable operating model.

Agenda:

- ArgoCD Overview
- Current Adoption
- Growth Opportunities
- Problem Statement
- Architecture Direction
- Feedback & Next Steps

Slide 2

What is ArgoCD?

GitOps Control Plane for Kubernetes

Key Capabilities:

- Git as Source of Truth
- Continuous Reconciliation
- Multi-Cluster Management
- Health & Visibility
- ApplicationSets
- RBAC & Multi-Tenancy

Why It Matters: Standardized deployment and operations platform for Kubernetes.

Slide 3

Managed ArgoCD Adoption Today

Platform Services

- Managed Spark Streaming
- Managed PostgreSQL

Application Teams

- API Proxy
- P13
- Other Single-Tenant Consumers

Value Delivered

- Standardized Deployments
- Visibility
- Automation
- RBAC
- Consistency

Slide 4

Why Expand Managed ArgoCD?

Opportunity for Broader Adoption

Potential Consumers:

- Namespace Services
- Large Cluster Fleets
- Shared Infrastructure Platforms

Benefits:

- GitOps Workflows
- Fleet Management
- Standardized Operations
- Visibility
- Self-Service

Opportunity: Reusable GitOps Platform Capability

Slide 5

Current Architecture Limits Future Adoption

Current Scale

• ~250 Clusters
• ~3,100 Applications

Current Scaling Status

• 14-node ArgoCD Cluster

Estimated Future Scale

• 2,000+ Clusters
• 200,000+ Applications

Problem Statement

• Shared Dependency
• Larger Blast Radius
• Centralized Scaling
• Increased Operational Risk

Key Observation Future adoption should not be constrained by a single control plane.

Slide 6

Solution Requirements & Value Proposition

Requirements

• Scalability
• Reliability
• Operational Simplicity
• Unified Experience

Value Proposition Enable Managed ArgoCD to become a reusable platform capability.

Success Criteria

• Broader Adoption
• Reliability
• Consistent User Experience
• Long-Term Growth

Slide 7

Proposed Architecture Direction

Distributed Control Plane Architecture

High-Level Approach Multiple ArgoCD Control Planes Unified Managed ArgoCD Experience

Principles

- Horizontal Scalability
- Reduced Blast Radius
- Operational Independence
- Unified Consumer Experience

Expected Outcome Scalable foundation for broader adoption.

Slide 8

Next Steps & Feedback Requested

Checkpoint 2

- Placement Strategy
- Routing
- Capacity Planning
- Migration Strategy
- Operational Model
- Reliability & DR

Feedback Requested

- Problem Validation
- Architecture Validation
- Risks & Constraints

Decision Requested Proceed with detailed architecture design.